

JOINT SESSION · WEEK 3

# Data workshop

Comparing Europe with Data. Be able to load and document the COIL data in your tool (Python, R, Gretl or Power BI) before the analysis starts.

Week 3 · 26 Oct – 1 Nov 2026 · 45 minutes · Microsoft Teams



# What we will do today



**5 min** The dataset: indicators, sources, coverage and gaps (especially Montenegro).

**20 min** Live demo, about 5 minutes per tool: loading the data, selecting countries and indicators and a first chart in Python, R, Gretl and Power BI.

**5 min** How to write the data dictionary and handle missing values.

**15 min** Questions from the teams.

# One row per country and year

| iso3 | country    | year | SL_UEM_1524_ZS | NY_GDP_PCAP_PP_KD | IT_NET_USER_ZS |
|------|------------|------|----------------|-------------------|----------------|
| ESP  | Spain      | 2022 | 29.7           | 46733             | 94.5           |
| ESP  | Spain      | 2023 | 28.7           | 47323             | 95.4           |
| SVK  | Slovakia   | 2022 | 19.9           | 38649             | 89.1           |
| SVK  | Slovakia   | 2023 | 19.9           | 39502             | 87.2           |
| MNE  | Montenegro | 2022 | 29.8           | 25583             | 88.2           |
| MNE  | Montenegro | 2023 | 23.3           | 27243             | 88.6           |

Cross-section = one year, many countries. Panel = many countries followed over many years. Codes are World Bank / IMF codes; in this file the dots are replaced by underscores, so the same names work in Python, R, Gretl and Power BI.

**Before any analysis, drop the row "EUU" (EU aggregate): it is not a country.**

# Three files with the same data

## **coil\_indicators\_gretl.csv**

One row per country and year; codes with underscores. Use it in Gretl, Python and R (all the code in the methods module uses this file).

## **coil\_indicators\_wide.csv**

One row per country and year; original codes with dots. Easiest for Excel and Power BI.

## **coil\_indicators\_long.csv**

One row per country, year and indicator, with unit and source in every row. Handy for the data dictionary.

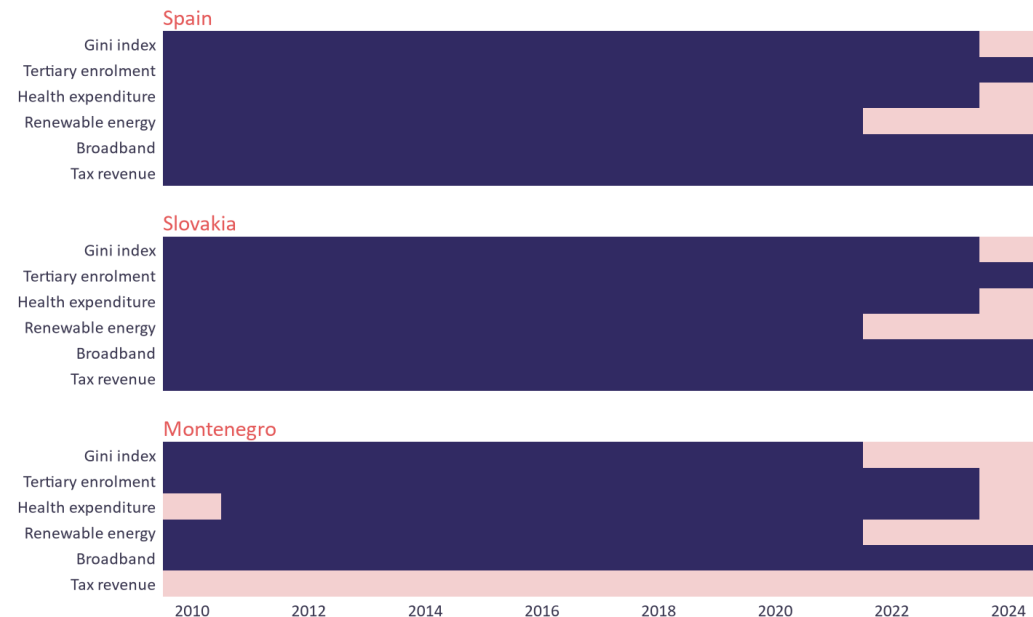
*The Data section also shows the coverage of every indicator by country, and the list of indicators with their units.*

# What is in the dataset

| Theme             | Indicators (unit)                                                                                                                                                                             |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Labour market     | Unemployment, total (% of labour force (modelled ILO estimate)) · Unemployment, youth (15-24) (% of labour force aged 15-24) · Female labour force participation (% of female population 15+) |
| Income and growth | GDP per capita, PPP (constant 2021 international \$) · GDP growth (annual %) · Inflation, consumer prices (annual %) · Services, value added (% of GDP)                                       |
| Inequality        | Gini index (index (0-100))                                                                                                                                                                    |
| Public finance    | General government gross debt (% of GDP) · Tax revenue (% of GDP)                                                                                                                             |
| Demography        | Population, total (persons) · Population aged 65 and above (% of total population) · Fertility rate (births per woman)                                                                        |
| Education         | Tertiary school enrolment (% gross) · Government expenditure on education (% of GDP)                                                                                                          |
| Health            | Life expectancy at birth (years) · Current health expenditure (% of GDP)                                                                                                                      |
| Digitalisation    | Individuals using the Internet (% of population) · Fixed broadband subscriptions (per 100 people)                                                                                             |
| Environment       | Renewable energy consumption (% of total final energy consumption) · CO2 emissions per capita (excl. LULUCF) (t CO2e per capita)                                                              |

## COVERAGE

# Gaps you have to know about

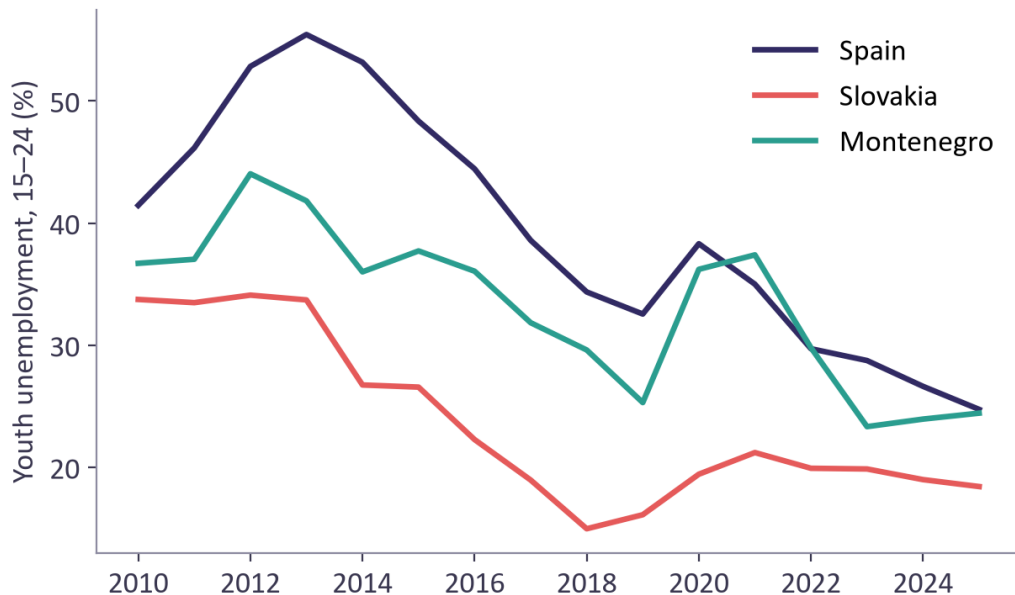


Dark = data available, pink = missing (2010–2024)

## What this means

- Montenegro has no tax revenue data (15 of 15 years missing), and no CO2 or education spending series.
- Renewable energy ends in 2021 for every country; the Gini index ends in 2022–2023.
- Health expenditure ends in 2023.
- Check coverage before you choose your indicators and years.

# The same first chart in four tools



## The task

- Load the COIL file.
- Keep Spain, Slovakia and Montenegro.
- Plot youth unemployment (SL\_UEM\_1524\_ZS) from 2010 to 2025.
- Each tool takes about 5 minutes: first without code (Power BI, Gretl), then with code (Python, R).

# Without writing code: Power BI and Gretl

## Power BI Desktop

- Get data > Text/CSV > coil\_indicators\_wide.csv > Load.
- Insert a Line chart.
- X-axis: year. Y-axis: SL.UEM.1524.ZS. Legend: country.
- Filters pane: country is Spain, Slovakia, Montenegro.
- Check that year is a whole number, not summarised.

## Gretl

- File > Open data > User file > coil\_indicators\_gretl.csv.
- Data > Dataset structure > Panel: unit = iso3, time = year.
- Sample > Restrict, based on criterion: iso3 == "ESP" || iso3 == "SVK" || iso3 == "MNE".
- For the chart, type one line in the Gretl console: `panplot SL_UEM_1524_ZS --overlay`

*Both routes give the chart on the previous slide. The step-by-step guide for each method is in the methods module.*

# With code: Python and R (if your course uses them)

## Python (pandas + matplotlib)

```
import pandas as pd
import matplotlib.pyplot as plt

d = pd.read_csv("coil_indicators_gretl.csv")
h = d[d.iso3.isin(["ESP", "SVK", "MNE"])]
h.pivot(index="year", columns="country",
         values="SL_UEM_1524_ZS").plot()
plt.ylabel("Youth unemployment, 15-24 (%)")
plt.show()
```

## R (ggplot2)

```
d <- read.csv("coil_indicators_gretl.csv")
h <- subset(d, iso3 %in% c("ESP", "SVK", "MNE"))

library(ggplot2)
ggplot(h, aes(year, SL_UEM_1524_ZS,
              colour = country)) +
  geom_line() +
  labs(y = "Youth unemployment, 15-24 (%)")
```

*Run the code from the folder where you saved the CSV file. In R, install ggplot2 once with `install.packages("ggplot2")`.*

# The data dictionary: one row per variable

| Variable    | Code              | Definition                  | Unit                    | Source     | Years     | Missing values and what we did                                                  |
|-------------|-------------------|-----------------------------|-------------------------|------------|-----------|---------------------------------------------------------------------------------|
| youth_unemp | SL.UEM.1524.ZS    | Unemployment, youth (15–24) | % of labour force 15–24 | World Bank | 2010–2025 | None                                                                            |
| health_exp  | SH.XPD.CHEX.GD.ZS | Current health expenditure  | % of GDP                | World Bank | 2010–2023 | Montenegro 2010 missing: Montenegro enters in 2011 (unbalanced panel)           |
| tax         | GC.TAX.TOTL.GD.ZS | Tax revenue                 | % of GDP                | World Bank | 2010–2024 | No data for Montenegro: country left out of this variable, stated in the report |

Write it as soon as you choose your indicators. It saves time in every later step and it is part of deliverable 2.

# Never fill a gap silently

## 1. Change the year or the indicator

Use a year or an indicator with full coverage for your countries.

## 2. Leave the country out

For that variable only, and say so in the report and in every table (n = ...).

## 3. Interpolate inside a country

Only between two known years, never beyond the last one.  
Mark the values you created.

## 4. Add another source

Eurostat or the national statistics office (e.g. MONSTAT).  
Document it in the data dictionary.

## Your checklist

☐ Your dataset is ready (one row per country and year) and every variable is documented in a data dictionary.

☐ Every variable has its code, unit, source and years in the data dictionary.

☐ You know which gaps affect your countries and what you will do with them.

☐ Week 4: exploratory data analysis. Descriptive statistics and first charts are done, and the team agrees on what the data show.

## Questions from the teams • 15 minutes